

# Experiment ‘p\_3\_4\_classification’ Results

June 3, 2026

**Experiment outcome:** SUCCESS

**Bad responses:** 0

**Responses containing assume :** 0

**Responses containing expect :** 0

**Resolution attempts:** 1

**Hard fails (resolution):** 0

**Soft fails (resolution):** 0

**Verification attempts:** 1

## Problem Specification

**Problem name:** p\_3\_4\_classification

**Natural language statement:** Write a method that takes three numbers and returns "all the same" if they are all the same, "all different" if they are all different, and "neither" otherwise.

**Method signature:** p\_3\_4\_classification (i: int, j: int, k: int) returns (s: string)

## Ensures

- if `i == j == k` then `s == "all the same"` else if `i != j && j != k && i != k` then `s == "all different"` else `s == "neither"`

## GenAI interactions

Below you will find all interactions between the ‘user’ (program) and the ‘assistant’ (OpenAI).

### Program → GenAI

You are given the following task to perform in Dafny:

Write a method that takes three numbers and returns "all the same" if they are all the same, "all different" if they are all different, and "neither" otherwise.

The signature should be:

`p_3_4_classification (i: int, j: int, k: int) returns (s: string)`

The method should respect the following contract:

`ensures if i == j == k then s == "all the same" else if i != j && j != k && i != k then s == "all different" else s == "neither"`

Produce and show only the Dafny body of this method, including the curly braces that surround it. Do not show the signature nor contract. You must not use 'assume' nor 'expect' anywhere in your code.

### GenAI → Program

**System fingerprint:** null

**ID:** chatcmpl-DmdpyOzQ7Y4QEtwlnQ2Wjn7VQPHaf

**Model:** gpt-5.5-2026-04-23

**Created at:** 1780485250

**Finish reason:** stop

**Usage:** 390 tokens in, and 214 tokens out

```
{
  if i == j && j == k {
    s := "all the same";
  } else if i != j && j != k && i != k {
    s := "all different";
  } else {
    s := "neither";
  }
}
```

## Final Program

```
method p_3_4_classification (i: int, j: int, k: int) returns (s: string)
  ensures if i == j == k then s == "all the same" else if i != j && j != k && i != k
    then s == "all different" else s == "neither"
{
  if i == j && j == k {
```

```
    s := "all the same";  
  } else if i != j && j != k && i != k {  
    s := "all different";  
  } else {  
    s := "neither";  
  }  
}
```

## Total Token Usage

Input tokens: 390

Output tokens: 214

Reasoning tokens: 130

Sum of 'total tokens': 604

## Experiment Timings

**Overall Experiment** started at 1780485249397, ended at 1780485258997, lasting 9600ms (9.60 seconds)

**Iteration #1** started at 1780485249404, ended at 1780485258996, lasting 9592ms (9.59 seconds)